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OpenEvidence Clinical Question-Answering Platform: Systematic Review of Early Evaluations

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Abstract

OpenEvidence is a retrieval-augmented clinical question-answering platform that provides responses grounded in curated biomedical sources with explicit citations and is reportedly used daily by over 40% of U.S. physicians. We systematically reviewed studies evaluating OpenEvidence for clinical question answering and decision support. MEDLINE/PubMed, Scopus, Web of Science, and Google Scholar were searched from inception to January 2026 for peer-reviewed studies conducted in clinical or clinically simulated settings. Eleven studies were included. OpenEvidence generally generated clinically relevant, evidence-supported responses and demonstrated lower rates of fabricated citations than general-purpose large language models in studies assessing hallucination. Performance was strongest in guideline-based settings but varied in complex clinical scenarios, and the platform often reinforced rather than altered clinical decisions. The evidence base remains limited by methodological heterogeneity, small sample sizes, and the platform's continuous evolution, highlighting the need for prospective real-world studies, standardized evaluation frameworks, and ongoing benchmarking.

Introduction

Clinicians frequently face time constraints when seeking accurate, up-to-date answers to complex clinical questions [1]. Traditional resources, such as clinical guidelines and primary literature, remain authoritative; however, they can be time-intensive to navigate [2]. These limitations have contributed to growing interest in conversational artificial intelligence systems that aim to synthesize medical evidence efficiently at the point of care [3, 4].

OpenEvidence is a clinical question-answering platform developed to address these challenges [5]. According to company-reported figures, the platform is consulted daily by over 40% of U.S. physicians [6].

Unlike general-purpose large language models (LLMs), OpenEvidence retrieves information from curated biomedical sources and provides explicit citations alongside generated responses [7]. The platform is intended to support evidence-informed clinical reasoning. Its development reflects broader interest in evidence-grounded AI in medicine, emphasizing transparency, traceability, and alignment with clinical standards [8].

Recent studies investigating OpenEvidence suggest that it can generate clinically relevant answers to real-world questions, often more efficiently than conventional information-seeking approaches [9]. However, the clinical application of OpenEvidence raises important methodological and practical concerns. The reliability of responses depends on the scope and quality of retrieved sources, as well as on how evidence is selected and summarized. The presence of citations does not ensure citation accuracy,

contextual relevance, or appropriate interpretation [10]. There is also concern that fluent and authoritative language may inadvertently over-influence clinicians, particularly in situations of diagnostic uncertainty or limited evidence [11]. These issues underscore the importance of a thorough evaluation of both performance and potential unintended consequences.

Existing studies evaluating OpenEvidence have explored diverse clinical applications and assessment approaches. However, the absence of a standardized evaluation framework for clinician-facing LLM systems makes comparison across studies challenging.. Differences in question formulation, comparator models, outcome metrics, and reference standards may influence reported performance. These methodological considerations accentuate the need for systematic synthesis and may inform future benchmarking strategies for evidence-grounded AI systems in medicine..

Given the scale of OpenEvidence usage by US physicians and the growing but fragmented body of evaluative research, a systematic review is warranted. To date, no systematic review has comprehensively examined the application of the platform for answering clinical questions and supporting clinical decision-making. This review aims to address this gap.

Results

Study Selection and Characteristics

Eleven studies published between 2024 and 2026 were included. All included studies evaluated OpenEvidence in clinical settings, most commonly through head-to-head comparisons with general-purpose or retrieval-augmented LLM systems. (**Table 1**).

Considerable methodological heterogeneity was observed across included studies. Evaluations differed in clinical domains, question sources, comparator systems, reference standards, and outcome definitions.

Some studies used real-world clinician-generated questions [13,15], whereas others relied on guideline-derived prompts, structured vignettes, or operational triage scenarios [20,24–26]. Comparator systems ranged from general-purpose LLMs to custom retrieval-augmented platforms and human administrative staff [16,19,21,26]. Reporting of AI-specific evaluation parameters such as testing dates, platform versions, prompting methodology, browsing capabilities, and repeat-run procedures was inconsistent across studies.

Study designs included retrospective case analyses (18%, 2/11), vignette-based evaluations (18%, 2/11), cross-sectional comparative studies comparing multiple AI systems on the same fixed set of clinical questions at a single time point (36%, 4/11),

and guideline-based benchmarking studies (27%, 3/11) (**Figure 3**). The number of clinical questions per study ranged from 15 to 130 and covered multiple clinical domains, including primary care, dentistry, cardiology, infectious diseases, ophthalmology, nutrition, gastroenterology, and clinical triage.

Expert clinician review served as the reference standard in all studies, with outcomes assessed using quantitative accuracy measures, Likert-scale ratings, or binary guideline adherence. Most studies compared OpenEvidence with general-purpose LLMs or competing retrieval-augmented language models, while a minority evaluated OpenEvidence in isolation .

Reported outcomes varied substantially across studies and included measures of accuracy or correctness, clinical utility, evidence grounding, hallucination or citation reliability, operational performance, and clinician-rated qualitative assessments. (**Table 2**).

Additional quantitative outcomes and methodological characteristics of individual studies are summarized in **Supplementary Table S3**.

Descriptive summary of results

General clinical question answering and decision support

Low et al. [13] evaluated OpenEvidence, and four other LLMs, on 50 clinician-generated clinical questions spanning multiple medical specialties. Retrieval-augmented and agentic systems substantially outperformed general-purpose LLMs in generating relevant, evidence-based, and clinically relevant responses. OpenEvidence produced clinically relevant answers for 48% of questions with existing evidence, whereas general-purpose models achieved clinically relevant responses in less than 5% of cases.

Hurt et al. [15] examined OpenEvidence in five real patient cases involving chronic conditions. Clinicians rated responses highly for relevance, clarity, and evidence support, although the impact on clinical decision-making was limited, suggesting that OpenEvidence primarily reinforced existing management plans rather than substantially altering them.

Collectively, these studies suggest that OpenEvidence performs best as a clinician-support tool for evidence retrieval and contextualization rather than as an autonomous decision-making system.

Specialty-specific evaluations across clinical domains

Several studies assessed OpenEvidence in specialty-focused clinical contexts, including dentistry, cardiology, infectious diseases, and ophthalmology.

Akpınar [16] compared ChatGPT 4.0 [17], OpenEvidence, and MediSearch [18] on frequently asked questions related to dental implants. Evaluations were made by oral and maxillofacial surgeons. Significant performance differences were observed

across models, with OpenEvidence performing favorably on several clinically oriented questions. In cardiology-focused evaluations, Hajj et al. [19] evaluated OpenEvidence and ChatGPT-4o using 15 clinician-facing questions related to transcatheter tricuspid valve interventions. They reported that ChatGPT-4o produced fully accurate answers more frequently than OpenEvidence (66.7% vs 26.7%) in transcatheter tricuspid valve intervention questions. However, both systems demonstrated incomplete or inaccurate responses in some scenarios.. Additional specialty-focused evaluations examined infectious diseases, ophthalmology, and implant dentistry contexts.

Borgonovo et al. [20] evaluated 15 LLMs using guideline-derived osteoarticular infection questions and reported that OpenEvidence and Microsoft Copilot achieved the highest overall accuracy (94.4%).

Khabaz et al. [21] assessed OpenEvidence in thyroid eye disease patient education questions. OpenEvidence achieved higher clinician-rated clinical scores than comparator systems (mean clinical score 5.96 vs 4.94 and 5.55, respectively; $p < 0.001$ for both comparisons), although ChatGPT demonstrated superior performance in non-clinical communication metrics (18.4 vs 15.0; $g = 1.25$; $p < 0.001$).

Hooshir [22] evaluated OpenEvidence and multiple GPT-based systems in implant dentistry. OpenEvidence and ScholarQA achieved 80% accuracy without fabricated citations, whereas general-purpose generative models demonstrated substantially higher hallucination rates. Taken together, Specialty-specific studies demonstrated strong performance of OpenEvidence in evidence-grounded and guideline-oriented tasks, while also highlighting variability across clinical domains and communication-focused evaluations.

Supportive clinical applications and patient-facing contexts

Hurt et al. [23] compared four LLMs in five clinical nutrition cases. Each system was queried with patient-specific questions. The blinded responses were rated by five nutrition experts. All models produced clinically appropriate responses, with Gemini achieving the highest mean scores for relevance (4.04) and clarity (4.16), and overall satisfaction scores ranging from 3.08 for Copilot to 3.84 for Gemini.

Guideline-based benchmarking and structured assessments

Masanneck et al. [24] evaluated OpenEvidence alongside several web-based retrieval-augmented systems using a 130-item guideline-derived question set. OpenEvidence achieved 82% accuracy, while domain whitelisting substantially improved the performance of competing web-based models. Toppeta et al. [25] evaluated multiple AI assistants using endoscopy guideline scenarios and observed substantial variability in model performance across time points, emphasizing the temporal instability of rapidly evolving LLM systems. Overall accuracies ranging from 50% to 90% in April and 63% to 80% in September. In September, both ChatGPT-4.5 and ChatGPT-4.0 achieved higher accuracy than participating physicians.

Overall, these studies underscore the importance of standardized benchmarking methodologies for evaluating clinician-facing AI systems across operational and guideline-based clinical settings.

Operational clinical decision support and triage

Wasserman et al. [26] compared the triage performance of five LLMs with that of human administrative staff, using 19 clinical vignettes. AI models demonstrated greater internal consistency (Cronbach's $\alpha = 0.950$). Among individual models, ChatGPT-o3 achieved the highest agreement with expert rankings (weighted $\kappa = 0.681$). These results suggest that contemporary LLMs align more closely with expert clinical judgment in patient prioritization than non-clinical staff.

Discussion

Across eleven reviewed studies, the current evidence base for OpenEvidence remains sparse relative to its widespread clinical adoption. Although these evaluations suggest that OpenEvidence generates clinically relevant and evidence-supported responses, often outperforming general-purpose LLMs in guideline-driven contexts [13,20,22], the findings are inherently limited by being snapshots taken at specific moments. Because OpenEvidence continuously evolves in its underlying model, retrieval mechanisms, and content sources, benchmarking at single time points significantly constrains the generalizability and comparability of these results.

An important finding is the lack of standardized methodologies for evaluating clinician-facing LLM systems. This heterogeneity complicates direct comparison across

studies and raises broader questions regarding how evidence-grounded AI systems should be benchmarked in real-world clinical practice.

Several studies demonstrated that OpenEvidence performs well in structured, guideline-driven settings. High accuracy rates were reported in benchmarking studies using validated guideline-derived questions, including osteoarticular infections and neurology guideline assessments, where OpenEvidence achieved accuracy rates exceeding 80% [20,24]. In studies specifically evaluating hallucination and citation reliability, retrieval-augmented systems such as OpenEvidence demonstrated lower rates of fabricated citations than several general-purpose generative models, although interpretive inaccuracies still occurred [22]. However, only a minority of included studies formally assessed citation fabrication or hallucination outcomes. These findings suggest that retrieval-based, curated evidence sources may offer safeguards against hallucination in clinical question answering.

At the same time, several studies reported that OpenEvidence primarily reinforced existing clinical decisions rather than altering management plans [15,23]. In primary care and nutrition case analyses, clinicians rated responses as accurate and relevant, but with limited impact on decision-making, indicating a supportive role rather than a transformative one [15,23]. In these studies, the platform's responses aligned with clinicians' initial reasoning and management strategies, serving to confirm or contextualize decisions rather than driving changes in care.

Although this pattern may reflect appropriate confirmatory decision support, it also raises concerns related to automation bias and LLM sycophancy, whereby fluent,

supportive outputs may increase user confidence without substantively improving decision quality, or when the LLM prioritizes agreeing with the user's beliefs or preferences over providing accurate or truthful information, respectively. [27, 28]. Similar findings were observed in patient education and triage scenarios, where OpenEvidence contributed to clinically appropriate guidance but did not replace expert judgment [21,26].

Comparative studies showed that OpenEvidence prioritizes evidence grounding and citation reliability, whereas other AI systems sometimes achieved higher accuracy or faster responses at the expense of increased hallucination risk [19,22]. In contrast, OpenEvidence demonstrated more consistent evidence grounding but lower accuracy in certain complex clinical scenarios, such as structural heart disease interventions [19].

The synthesized findings also have important implications for clinicians currently using OpenEvidence and related AI systems in practice. Across studies, OpenEvidence generally performed best when functioning as a supportive evidence retrieval and contextualization tool rather than as an autonomous clinical reasoning system. Although retrieval grounding may reduce hallucination and improve citation reliability, several studies demonstrated persistent interpretive inaccuracies, incomplete responses, and variability across clinical domains [19,21,22]. Clinicians should therefore interpret outputs cautiously and continue to rely on independent clinical judgment.

In addition, retrieval-augmented systems may selectively prioritize certain studies, journals, or evidence sources over others through ranking and retrieval mechanisms that remain incompletely transparent to users. Medical evidence is often

heterogeneous or conflicting, limited auditability regarding why specific sources were retrieved or emphasized may introduce additional challenges for interpretation and clinical trust [29].

Across studies, several recurring limitations and risks were identified (**Table 3**). These included dependence on the availability and quality of existing evidence [13], variable performance across clinical domains and question types [16,20], limited influence on clinical decision-making [15], and persistent interpretive errors despite accurate citation practices [22].

Additional concerns included lower performance on non-clinical communication metrics, such as readability or empathy [30, 31], in patient-facing contexts [18], and temporal variability in performance across repeated evaluations [25]. Importantly, hallucinations and incorrect interpretations were observed even in the absence of fabricated citations, underscoring that evidence grounding alone does not eliminate clinical risk.

Several limitations of both the current evidence base and this review should be acknowledged. Database searches required “OpenEvidence” to appear within indexed title or abstract fields; therefore, some potentially relevant studies mentioning the platform only within the full text or supplementary materials may not have been identified. In order to mitigate this limitation, manual citation tracking, reference screening, and database “similar articles” functions were additionally reviewed.

In addition, we included only peer-reviewed original research articles and excluded preprints, conference proceedings, technical reports, and benchmark repositories. Although this approach was intended to prioritize methodological rigor and clinically vetted evidence, it may have excluded emerging evaluations in the rapidly evolving generative AI literature.

Study designs were heterogeneous, outcome measures were inconsistently reported, and sample sizes were often small. Most evaluations relied on expert clinician judgment rather than downstream clinical outcomes, and few studies assessed real-world longitudinal use [13, 15–16, 19–26]. An additional limitation is that OpenEvidence is not a static system. Like other LLMs, it is continuously updated, including changes to underlying models, retrieval indices, and system interfaces. As a result, performance characteristics observed in individual studies may not persist over time, limiting the generalizability of point-in-time evaluations [32]. This temporal variability complicates direct comparison across studies conducted at different periods and underscores the need for caution when interpreting aggregated results.

The dynamic nature of modern AI platforms raises an important methodological question regarding whether static benchmarking approaches are sufficient for continuously updated clinical AI systems. Future research may therefore require continuously updated benchmarking frameworks, prospective real-world evaluations, and standardized clinically meaningful endpoints capable of adapting alongside evolving AI technologies.

An important limitation is the substantial methodological heterogeneity across studies evaluating OpenEvidence [13-26]. Included studies varied considerably in clinical domains, question sources, comparator models, scoring systems, and definitions of accuracy or clinical utility. Some studies used real-world clinician-generated questions [13,15], whereas others relied on guideline-derived prompts or structured vignettes [20,24,25]. This variability complicates direct comparison across studies and limits the ability to establish performance benchmarks.

Importantly, several of the most clinically relevant observations identified, including reinforcement of existing clinician reasoning, temporal variability, and benchmarking inconsistency, were qualitative rather than directly quantifiable findings.

Furthermore, most included studies did not explicitly report whether external web search or browsing capabilities were enabled for general-purpose models such as ChatGPT, limiting reproducibility and potentially affecting comparability with retrieval-augmented systems [16,19,21–23,25,26].

A considerable limitation of the current evidence base is the minimal evaluation of security, privacy, governance, and infrastructure-level risks specific to retrieval-augmented clinical AI systems [33]. Most included studies focused primarily on comparative performance metrics, clinical relevance, or hallucination reduction rather than broader operational safety considerations.

However, deployment of clinician-facing RAG systems raises important unresolved concerns regarding data poisoning attacks within retrieval corpora, indirect prompt injection through retrieved sources, inclusion of manipulated or fabricated documents, and governance of continuously evolving indexed evidence repositories. Additional challenges include retrieval traceability and auditability, corpus provenance and potential privacy risks involving retrieval logs, search histories, or vector databases [33].

These concerns may be particularly important in closed commercial systems where retrieval infrastructure, indexing methodology, and corpus governance remain incompletely transparent [34]. Furthermore, reduced hallucination through retrieval grounding should not be interpreted as equivalent to overall system safety or clinical correctness, as evidence-linked outputs may still contain interpretive inaccuracies or misleading recommendations despite accurate citations [22]. Future evaluations of clinical AI systems should include broader security, governance, and regulatory assessment frameworks.

Greater attention is needed to understand how clinicians integrate OpenEvidence into decision-making workflows and how sustained use may influence clinical reasoning, trust, and reliance over time. In parallel, deployment efforts should include clinicians' education about the potential benefits as well as the persistent risks and pitfalls of using these systems, even if grounded in high quality retrieval index. As evidence-grounded

AI platforms continue to be adopted, thorough evaluation and transparent reporting will remain essential to support safe and effective integration into clinical practice.

In conclusion, the rapid adoption of OpenEvidence among clinicians outpaces the sparse research available. With just 11 studies, limited in scope or confined to niche clinical domains, the evidence is thin relative to its purported usage in daily medical practice. Adoption of such platforms demands prospective real-world studies with clinical endpoints and ongoing benchmarking.

Methods

Literature Search

A systematic literature search was conducted to identify studies evaluating OpenEvidence for answering clinical questions and supporting clinical decision-making. The search was performed in MEDLINE/PubMed, Scopus, Web of Science, Cochrane Library, and Google Scholar. Databases were searched from inception to January 2026. No restrictions were applied based on clinical specialty. “OpenEvidence” is a highly

specific and relatively recent platform; hence, database retrieval yielded a limited number of directly relevant records despite broad database coverage.

Online-ahead-of-print articles were eligible if electronically indexed and accessible prior to the final search date. Preprints, conference abstracts without full text, company webpages, news releases, and other grey literature sources were excluded from formal study inclusion but could be cited separately for contextual background information.

The search strategy was designed to capture empirical evaluations of the OpenEvidence platform in clinical contexts. Full database-specific search strategies, search dates, and supplementary search procedures are provided in **Supplementary Material**.

Reference lists of included articles were manually screened to identify additional relevant studies. Citation tracking and database-generated “similar articles” features were also reviewed to ensure comprehensive coverage.

Ethical approval was not required, as this study synthesizes previously published literature and does not involve the collection of primary data or the disclosure of identifiable participant information. The review was conducted in accordance with the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines [12]. The study protocol was registered with PROSPERO (CRD420261289103).

Study Selection

All records retrieved from the database searches were imported into a reference management system, and duplicates were removed. Titles and abstracts were independently screened by two reviewers (YA and VS). Full-text articles were independently assessed by two reviewers (EK and BSG). Data extraction and risk-of-bias assessment were independently performed by two reviewers (YA and VS) using standardized forms. Disagreements at all stages were resolved through discussion and consensus among the review team. . The study selection process is summarized in

Figure 1.

Inclusion Criteria

Studies were eligible for inclusion if they met all of the following criteria:

(1) Evaluated the OpenEvidence platform for answering clinical questions or supporting clinical decision-making. (2) Examined real or simulated clinical questions. (3) Reported empirical outcomes. (4) Involved clinical contexts, including clinician-facing evaluations, patient-derived clinical scenarios, guideline-based questions, or multiple-choice questions designed to assess clinically relevant knowledge. (5) Published in English. (6) Peer-reviewed original research articles.

Exclusion Criteria

Studies were excluded if they met any of the following criteria:

(1) Did not evaluate the OpenEvidence platform. (2) Focused on other retrieval-augmented generation systems, without reference to OpenEvidence. (3) Addressed non-clinical tasks. (4) Conducted exclusively in non-clinical technical settings without clinician involvement or guideline-based context. (5) Editorials, commentaries, narrative reviews, protocols, conference abstracts, or non-peer-reviewed publications. (6) Published in languages other than English.

Risk of Bias Assessment

The risk of bias and applicability concerns of included studies were independently assessed by two reviewers using an adapted version of the Quality Assessment of Diagnostic Accuracy Studies 2 (QUADAS-2) tool, modified to reflect the evaluation of AI-based clinical decision-support platforms. Detailed justifications for QUADAS-2 risk of bias and applicability judgments for each included study are provided in **Supplementary Table S1**.

Although QUADAS-2 was adapted to assess studies evaluating AI-based clinical decision-support tools, some domains do not fully capture sources of bias unique to LLMs, and unclear ratings primarily reflect limitations of reporting frameworks rather than study quality.

We retained adapted QUADAS-2 as the primary risk-of-bias tool because it is widely established, and maps transparently onto the core structure of the included

comparative evaluation studies. In parallel, we conducted an additional QUADAS-AI-informed appraisal detailed in **Supplementary Figure S1 and Supplementary Table S2**, to capture AI-specific sources of bias that are not consistently addressed by QUADAS-2.

The supplementary QUADAS-AI-informed appraisal identified several recurring AI-specific methodological limitations across studies. Common concerns included incomplete reporting of prompting methodology, model configuration, platform versioning, temporal testing conditions, and reproducibility procedures. Many studies also lacked detailed information regarding browsing capabilities and repeat-query evaluation, limiting reproducibility and longitudinal comparability across continuously evolving clinician-facing AI systems.

QUADAS-AI judgments were based on full-text methods when available. For studies without full-text access, ratings were based on reported abstract-level information and were marked unclear when details were insufficient. Disagreements were resolved through consensus. Results of the quality assessment are presented in **Figure 2**.

Declarations

(1) Availability of data and materials:

All data generated or analyzed during this study are included in this published article and its supplementary information files.

(2) Acknowledgements:

No funding was received for this research

(3) Authors' contributions:

(4) YA and EK conceived the study, YA drafted the manuscript, YA and VS designed the review protocol, independently conducted the literature search, study selection, data extraction, and risk-of-bias assessment and performed the data synthesis. EK and BSG independently assessed full-text articles for eligibility and resolved disagreements through discussion with the review team when necessary. RF, PK, AKB, and GNN contributed to the interpretation of the findings and critically revised the manuscript for important intellectual content. All authors read and approved the final manuscript.

(5) Competing interests:

The authors declare no competing financial or non-financial interests.

(6) Ethics approval and consent to participate:

Not applicable

(7) Consent for publication:

Not applicable.

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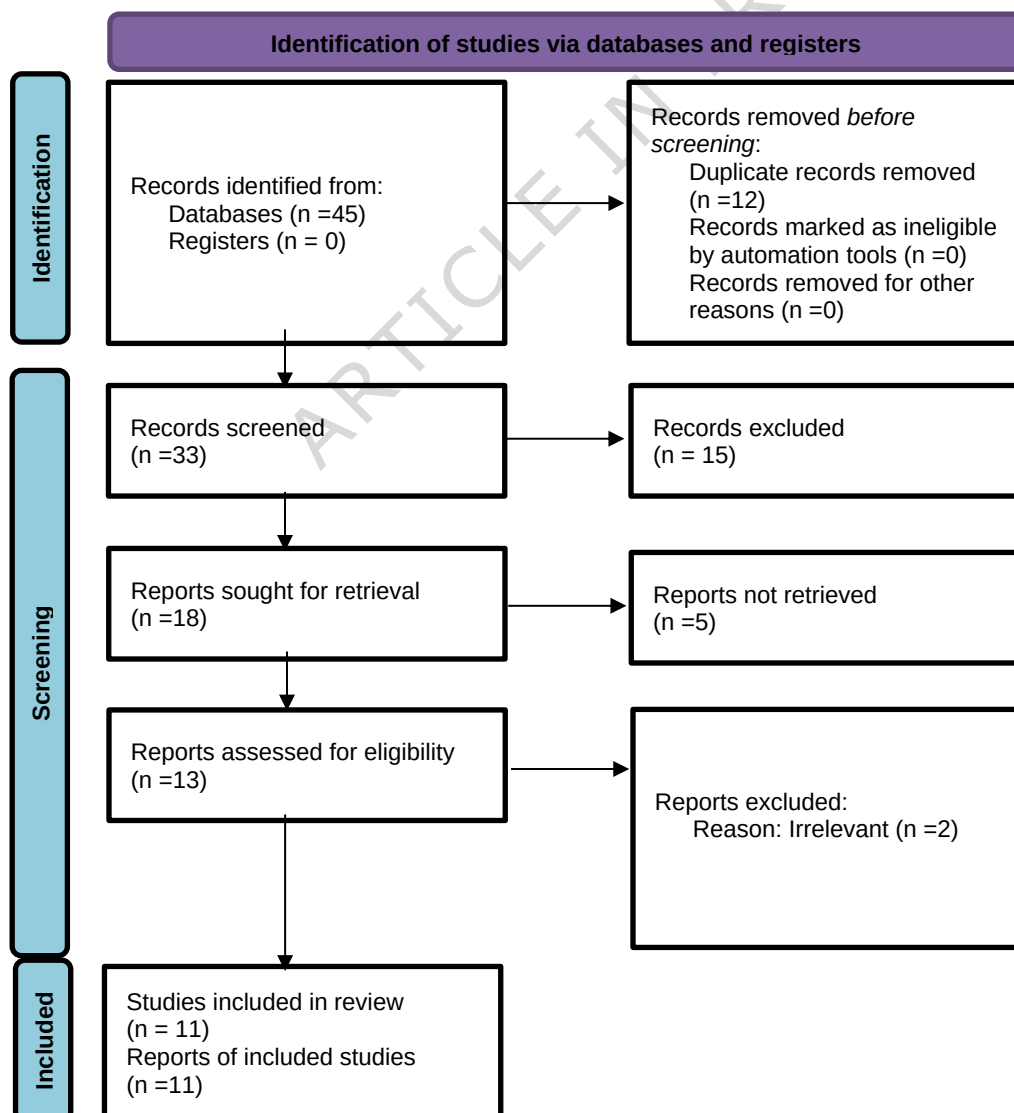
Figure 1. Flow diagram of inclusion and exclusion process

Figure 2. Risk of bias and applicability concerns using UADAS-2 tool

	RISK OF BIAS				APPLICABILITY CONCERNS		
	PATIENT SELECTI ON	INDE X TEST	REFEREN CE STANDAR D	FLOW AND TIMIN G	PATIENT SELECTI ON	INDE X TEST	REFEREN CE STANDAR D
Low al.	?	?	?	?	■	■	■
Hurt et al.	■	?	?	?	■	■	■
Akpınar et al.	?	?	?	?	?	■	■
Hajj et al.	■	?	▲	?	?	■	■
Borgonovo et al.	■	■	■	■	▲	■	■
Khabaz et al.	?	?	?	?	▲	■	?
Hooshlar et al.	■	?	?	?	■	■	?
Hurt et al.	▲	?	■	?	▲	■	■
Masanneck et al.	■	■	■	■	■	■	■
Toppeta et al.	■	?	■	■	■	■	■
Wasserman et al.	?	?	■	■	■	■	■

Low risk ■
High risk ▲
Unclear ?

Figure 3. Schematic representation of Study Selection and Characteristics

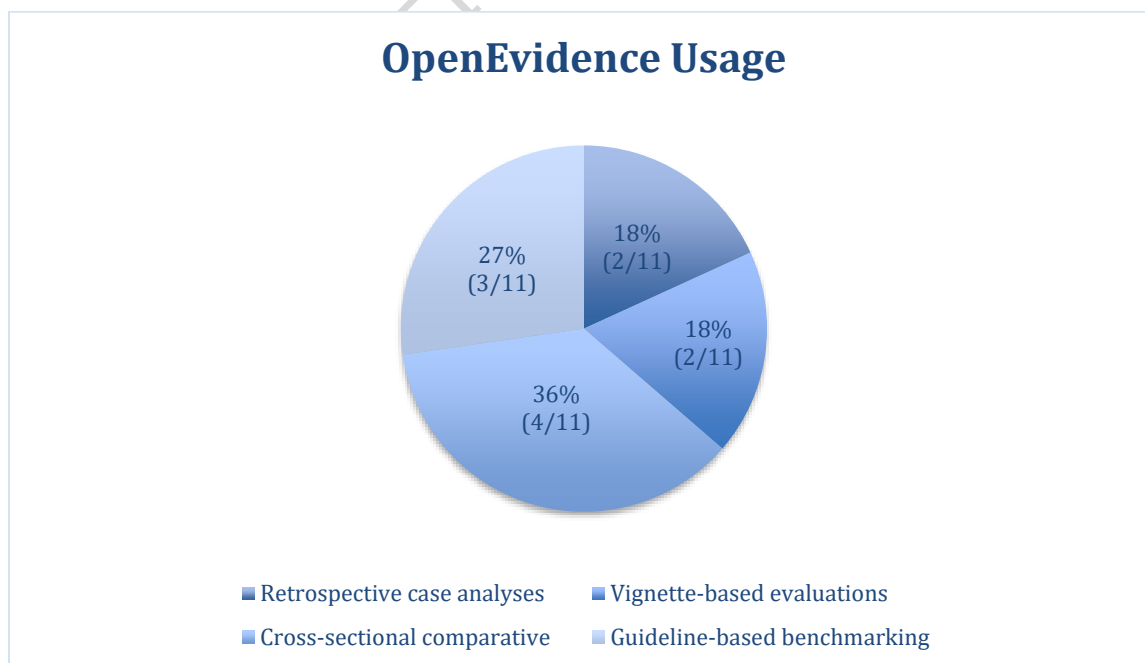


Table 1. Characteristics of Included Studies Evaluating the OpenEvidence Platform

Study	Year	Clinical Domain	Study Design	Number of questions / cases	Comparator(s)	Primary outcomes assessed
Low et al. [13]	2024	General clinical decision-making	Cross-sectional comparative	50 clinical questions	General-purpose LLMs; ChatRWD	Accuracy, evidence relevance
Hurt et al. [15]	2024	Primary care (chronic disease)	Retrospective case analysis	5 patient cases	None	Clarity, relevance, evidence support, decision impact
Akpınar [16]	2024	Dentistry (implants)	Cross-sectional comparative	20 questions	ChatGPT-4.0; MediSearch	Accuracy, clinical relevance

Hajj et al. ^[19]	2024	Cardiology (structural heart disease)	Cross-sectional comparative	15 clinician-facing questions	ChatGPT-4o	Accuracy, response time
Borgonovo et al. ^[20]	2025	Infectious diseases (osteoarticular infections)	Guideline-based benchmarking	126 guideline-based questions	Multiple LLMs (n = 14)	Guideline adherence, accuracy
Khabaz et al. ^[21]	2025	Ophthalmology (thyroid eye disease)	Cross-sectional comparative	15 patient education questions	GPT-4o-mini; RAG model	Clinical accuracy, communication quality
Hooshiair ^[22]	2025	Dentistry (implantology)	Cross-sectional comparative	15 prompts	GPT-4o; ScholarQA	Accuracy, hallucination, citation fabrication
Hurt et al. ^[23]	2025	Clinical nutrition	Retrospective case analysis	5 patient cases	ChatGPT; Gemini; Copilot	Clinical appropriateness, satisfaction
Masanneck et al. ^[24]	2025	Neurology (guideline QA)	Guideline-based benchmarking	130 questions	Perplexity Sonar variants	Accuracy, source quality
Toppeta et al. ^[25]	2025	Gastroenterology (endoscopy)	Vignette-based evaluation	Not specified (guideline vignettes)	ChatGPT-4.0; ChatGPT-4.5; Gemini; Claude	Appropriateness, guideline adherence
Wasserman et al. ^[26]	2025	Clinical triage	Cross-sectional comparative	19 vignettes	Other LLMs; human staff	Agreement with expert ranking

Table 2. Quantitative Outcomes Reported in Studies Evaluating the OpenEvidence Platform

Study	Clinical Context	Main Findings
Low et al. ^[13]	Real-world clinician-generated questions across multiple specialties	OpenEvidence produced clinically relevant evidence-supported responses for 48% of questions with existing evidence, compared with <5% for general-purpose LLMs
Hurt et al. ^[15]	Primary care chronic disease management	Mean clinician ratings were high for relevance (3.75 ± 0.44) and overall satisfaction (3.60 ± 0.60)

Akpınar ^[16]	Dental implant questions	OpenEvidence performed favorably on several implant dentistry questions compared with ChatGPT-4.0 and MediSearch Significant differences among models ($p < 0.01$)
Hajj et al. ^[19]	Structural heart disease interventions	ChatGPT-4o achieved fully accurate responses in 66.7% of cases vs. 26.7% for OpenEvidence
Borgonovo et al. ^[20]	Osteoarticular infection guideline questions	OpenEvidence achieved 94.4% accuracy (119/126 correct answers), among the highest-performing systems
Khabaz et al. ^[21]	Thyroid eye disease patient education	OpenEvidence achieved higher clinician-rated clinical scores than comparator models (mean score 5.96)
Hooshiair ^[22]	Implant dentistry prompts	OpenEvidence and ScholarQA achieved 80% accuracy without fabricated citations
Hurt et al. ^[23]	Clinical nutrition patient scenarios	OpenEvidence generated clinically appropriate responses across all nutrition cases with overall satisfaction ~3.5–3.8
Masanneck et al. ^[24]	Neurology guideline question answering	OpenEvidence achieved 82% accuracy in guideline-based question answering
Toppeta et al. ^[25]	Endoscopy guideline recommendations	AI model performance ranged from 50–90% in April and 63–80% in September evaluations
Wasserman et al. ^[26]	Clinical triage scenarios	AI systems demonstrated higher agreement with expert triage rankings ($\kappa = 0.572$) than administrative staff

Table 3. Limitations and Risks of the OpenEvidence Platform Identified Across Included Studies

Study	Type of Limitation or Risk	Description of Reported Limitation	Context
Low et al. ^[13]	Evidence dependence	Reduced actionability when relevant literature was absent	Real-world clinical questions

Hurt et al. ^[15]	Limited decision impact	Responses primarily reinforced existing clinical plans rather than altering decisions	Primary care chronic disease cases
Akpınar ^[16]	Variable accuracy	Performance varied across question types and individual items	Dental implant questions
Hajj et al. ^[19]	Lower accuracy in complex cases	Higher rate of inaccurate responses compared with a general-purpose model	Structural heart disease decision support
Borgonovo et al. ^[20]	Category-specific variability	Performance differed across infection categories and task types	Guideline-based infectious disease cases
Khabaz et al. ^[21]	Communication limitations	Lower scores on non-clinical metrics such as empathy and readability	Patient education in ophthalmology
Hooshlar ^[22]	Interpretation errors	Incorrect interpretations occurred despite absence of fabricated citations	Implant dentistry prompts
Hurt et al. ^[23]	Limited originality	Responses showed limited novelty and variable citation quality	Clinical nutrition cases
Masanneck et al. ^[24]	Source rigidity	Accuracy was not influenced by source type, unlike web-based models	Neurology guideline question answering
Toppeta et al. ^[25]	Temporal variability	Performance varied across time points and prompts	Endoscopy guideline recommendations
Wasserman et al. ^[26]	Role limitation	Performance did not replace need for clinical oversight	Clinical triage scenarios